TalentAP Mega Hackathon 2026: Data Guide

Retail · Healthcare · BFSI · Data version **v1.0**

This guide tells you **which file to download** and **what to do with it, step by step**. Please read it once from top to bottom before you start. It takes about 10 minutes.

First, the 5 most important points

1. **Download only the zip file for your own problem statement.** Each problem statement has its own zip file.
2. **All the data is synthetic.** A computer program created it for this hackathon. There are no real customers, patients or bank users in it.
3. **After you unzip, open the `README.md` file inside your folder.** That file is the full guide for your problem. It explains every file and every column. For Retail and Healthcare, it also explains what to submit and how you are scored.
4. **Retail and Healthcare:** the answers (labels) for the test data are **hidden on purpose**. You will not find them in the zip. The organisers keep them and use them to score your work.
5. **BFSI:** the zip gives you **data only**. Your task and expected outcomes are **exactly as written in the published problem statement**. Nothing has been added or changed.

Step 1: Download the zip for your problem statement

Find your problem statement ID (for example **R2** or **H3**) in the table and download that one file.

Domain	ID	Problem statement	File to download	Size
Retail	**R1**	The Phantom Stock-Out Detector	`R1_Phantom_StockOut_Detector_v1.0.zip`	14 MB
Retail	**R2**	The Returns Fraud vs. Genuine Grievance Sorter	`R2_Returns_Fraud_vs_Genuine_Grievance_v1.0.zip`	5 MB
Retail	**R3**	The Conversational Shopping Assistant That Doesn't Lie	`R3_Grounded_Shopping_Assistant_v1.0.zip`	0.9 MB
Retail	**R4**	Demand Forecasting for the New Product With No History	`R4_New_Product_Demand_Forecasting_v1.0.zip`	1.4 MB
Retail	**R5**	Turning Store Chatter Into Action: The Voice-of-Store Engine	`R5_Voice_of_Store_Engine_v1.0.zip`	2.7 MB
Healthcare	**H1**	OPD Triage Assistant for an Overloaded Government Hospital	`H1_OPD_Triage_Assistant_v1.0.zip`	2.2 MB
Healthcare	**H2**	Discharge Summary to Patient-Understandable Care Plan	`H2_Discharge_Summary_to_Care_Plan_v1.0.zip`	5 MB

Domain	ID	Problem statement	File to download	Size
Healthcare	**H3**	Predicting the 30-Day Readmission — And Doing Something About It	`H3_30Day_Readmission_Risk_v1.0.zip`	1.6 MB
Healthcare	**H4**	The Medication Safety Second Pair of Eyes	`H4_Medication_Safety_Checker_v1.0.zip`	0.9 MB
Healthcare	**H5**	Frontline Health-Worker Companion for Low-Connectivity Areas	`H5_Frontline_Health_Worker_Companion_v1.0.zip`	0.2 MB
BFSI	**B1**	Smart Data Masking System for PII Protection (BFSI Problem Statement 1)	`B1_Smart_Data_Masking_PII_v1.0.zip`	39 MB
BFSI	**B2**	Intelligent Life Insurance Needs Finder & Digital Advisory Platform (BFSI Problem Statement 2)	`B2_Life_Insurance_Needs_Finder_v1.0.zip`	0.3 MB

Optional: check that your download is not damaged.

1. Also download `SHA256SUMS.txt` from the same page. It lists a code (a "checksum") for every zip.
2. Open a terminal in the folder that has your zip, usually **Downloads** (Step 5 shows how to open a terminal in a folder).
3. Run the command for your computer. Use your own zip's name in place of the R2 name.
   - Windows: `certutil -hashfile R2_Returns_Fraud_vs_Genuine_Grievance_v1.0.zip SHA256`
   - Mac: `shasum -a 256 R2_Returns_Fraud_vs_Genuine_Grievance_v1.0.zip`
   - Linux: `sha256sum R2_Returns_Fraud_vs_Genuine_Grievance_v1.0.zip`
4. Find your zip's line in `SHA256SUMS.txt`. If the two codes match, the file is fine. If they do not match, download the zip again.

Step 2: Unzip the file

- **Windows:** right-click the zip → **Extract All…** → **Extract**.
- **Mac:** double-click the zip.
- **Linux:** run `unzip R2_Returns_Fraud_vs_Genuine_Grievance_v1.0.zip` in a terminal (use your own zip's name).

Important (Windows, WinRAR and 7-Zip): these usually put the folder inside one more folder with almost the same name, like this:

```
R2_Returns_Fraud_vs_Genuine_Grievance_v1.0\
└── R2_Returns_Fraud_vs_Genuine_Grievance\    ← use THIS inner folder
    ├── README.md
    └── ...
```

Always work in **the folder that directly contains** `README.md`. In the rest of this guide we call it **"your problem folder"**.

What you will find in your problem folder

Retail (R1 to R5) and Healthcare (H1 to H5):

```
your problem folder/
├── README.md          ← START HERE: the full guide for your problem
├── data/              training data (answers given) + test data (answers hidden)
├── sample_submission  shows the correct FORMAT only (the answers in it are useless)
├── evaluate.py        the scoring program (the organisers use the same one)
└── baseline/          a simple starting solution that already works
```

The sample submission is one or two files, or a folder for some problems. Its answers are random or deliberately poor. Use it only to see the format.

R3 and H5 also have `MODELS.md` and `download_models.py`. These help you download small AI models that work offline. Using them is optional. If you want them, **download them before the event on good internet**, because they are large. `MODELS.md` shows how.

BFSI (B1 and B2): the folder has `README.md` and a `data/` folder.

- **B1** also has ready-to-use database files (`sql/`), a loading script (`load/`) and a `docker-compose.yml`.
- **B2** also has a glossary of insurance words (`docs/glossary.md`).

BFSI zips do **not** have `evaluate.py`, a sample submission or a baseline. This is on purpose, because the published BFSI problem statements do not ask for them.

Step 3: Read your problem's README.md

Open `README.md` in **VS Code** (press `Ctrl+Shift+V` on Windows, `Cmd+Shift+V` on Mac to see it nicely formatted) or in any text editor (Notepad on Windows, TextEdit on Mac).

Read these parts carefully:

Part of your README	What it tells you
The top part (before the first heading)	The story and what you must build and show
What's in the box	Every file and how many rows it has
Data dictionary (B1: **Tables and columns**)	What every column means
Know your data	Honest warnings: messy text, missing values, typing mistakes
What to submit (Retail and Healthcare)	The exact file names and columns you must submit, and special rules
How you're scored (Retail and Healthcare)	How `evaluate.py` gives your score
What else we expect (H3 only)	Extra things H3 teams must show
Hour Zero quick start (B1: **Loading the data**)	The first things to run

Part of your README	What it tells you
Assumptions you may make (Retail and Healthcare)	What you are allowed to assume

Step 4: Set up Python (one time only)

You need **Python 3.9 or newer**. (3.10, 3.11, 3.12 and 3.13 are all newer than 3.9.)

4a. Check if Python is installed. Open a terminal (see Step 5 if you do not know how) and type:

Windows	Mac or Linux
`python --version`	`python3 --version`

If you see `Python 3.9` or a higher number (for example `Python 3.12.4`), go to 4b.

If you see an error, or a version lower than 3.9, install Python from **https://www.python.org/downloads/**. **Windows users:** on the first screen of the installer, **tick the box "Add python.exe to PATH"** before you click Install.

On a new Mac, a window may ask you to install "command line developer tools". Click **Install**, wait for it to finish, then type the command again.

4b. Install the libraries. Most teams need `pandas` and `numpy`:

Windows	Mac or Linux
`python -m pip install pandas numpy`	`python3 -m pip install pandas numpy`

R1 teams only also need `pyarrow`:

Windows	Mac or Linux
`python -m pip install pyarrow`	`python3 -m pip install pyarrow`

If your README lists any other library, install it the same way.

> **Note for Mac and Linux:** the READMEs inside the zips mostly write `python`. On Mac and Linux, type `python3` wherever you see `python`.

Step 5: Open a terminal inside your problem folder

All commands must be run **from inside your problem folder**. If you run them from somewhere else, you will get "file not found" errors.

- **Windows:** open your problem folder in File Explorer. Click the address bar at the top, type `cmd` and press **Enter**. A black Command Prompt window opens, already inside the folder.
- **Mac:** open the **Terminal** app. Type `cd` followed by a space, drag your problem folder from Finder into the Terminal window, and press **Enter**.
- **VS Code (any computer): File → Open Folder…** → choose your problem folder. Then **Terminal → New Terminal**.

Check that you are in the right place. Type `dir` (Windows) or `ls` (Mac or Linux) and press Enter. You should see `README.md` and `data` in the list. If you do not, you are in the wrong folder.

Step 6: Run your first commands ("Hour Zero")

Retail and Healthcare: run the baseline and score it

Go to the **Hour Zero quick start** section of your README. Copy the commands **one at a time**, paste them into the terminal, and press Enter after each one.

For example, these are the commands for **R2**:

```
python baseline/baseline.py
python evaluate.py --submission submission/R2_predictions.csv --truth
data/return_requests_train.csv
python -m json.tool score.json
```

What each command does (for R2; the file names in your README may be different):

1. `baseline/baseline.py` runs the simple starting solution. It writes prediction files.
2. `evaluate.py` compares those predictions with the answers you **do** have (training or dev data). It prints a report on the screen and also saves it in a `.json` file (usually `score.json`).
3. `json.tool` shows that `.json` file in a readable way.

Your README also says which command makes your **test** submission file. Run that too.

These commands work as written on Windows, Mac and Linux (remember `python3` on Mac and Linux).

How do you know it worked? In the report, find the line that starts with **PRIMARY** (for R1 it says `PRIMARY precision@20`). That is your main score. **Higher is better.** If your number is close to the one in this table, your setup is correct:

ID	Baseline primary score in Hour Zero	Scored on
R1	about 0.40	dev data
R2	about 0.03	training data
R3	about 0.73	dev data
R4	about 0.63	dev data
R5	about 0.48	training data
H1	about 0.46	training data
H2	about 0.52	dev data
H3	about 0.44	training data
H4	about 0.41	training data
H5	about 0.31	dev data

Your README may also give the baseline's score on the hidden test data. That number can be different, because it is measured on different data. To check your setup, compare with the table above.

The baseline is simple on purpose. Your job is to build something much better. The score is also **not** plain accuracy: understanding how the score works is part of the challenge, so read **How you're scored** in your README.

BFSI

- **B1:** the easiest way is the ready SQLite database. It needs no installation; follow **Option 1** in the **Loading the data** section of your README. PostgreSQL and MySQL with Docker (Option 2) are also available. **If you plan to use Docker, download the Docker images before the event on a good internet connection** (about 1.3 GB; the README shows the command).
- **B2:** your README's **Hour Zero quick start** section has ready Python code, not terminal commands. Copy that code into a new file named `start.py` inside your problem folder. Then run `python start.py` (Mac and Linux: `python3 start.py`). It should print `1.044`.

Then explore the data and build what your **published problem statement** asks for.

Step 7: Build your solution

- **Check your work on the training or dev data,** using `evaluate.py` (Retail and Healthcare). You will not see the test answers.
- **Read the rules in the "What to submit" section of your problem's README.** Some problems have special rules. For example, **R1** and **R5** must not use "future" data. The organisers check this from your outputs and code, and may ask you to **run your code in front of the jury** on a cut-off (R1) or disguised (R5) copy of the data. **H4** predictions must come from your program, not from labelling patients by hand or one by one in a chatbot.
- **Keep your code tidy and runnable, and follow the *Participant Coding Guidelines*.** Every team's code is reviewed, and the judges look at more than the score (see "How you are judged" below).

Step 8: Submit (all teams)

Every team submits **one zip of its whole project**: **all** your code, your own `README.md`, and (Retail and Healthcare only) your prediction files. The *Participant Coding Guidelines* explain how to write and package your code. Read them too.

1. Your zip file must be named `teamID_psID.zip`. Use the team ID the organisers gave you. For example, team **T042** working on **H3** submits `T042_H3.zip`.

2. What goes in the zip:

```
T042_H3.zip
└── T042_H3/                    your project (a fresh git clone, made in step 4)
    ├── README.md               YOUR README: how to set up and run your code
    ├── submission/             Retail and Healthcare only
    │   └── H3_predictions.csv
    └── FULL_CODE/              This folder should contain all of your code:
                                your demo app + the code that created the
                                submission/ files
```

- **All your code, inside a folder named exactly `FULL_CODE/`, and your own `README.md`.** This is required for **every** team, including BFSI. "All your code" means the code of the solution you demo (for example a web or mobile app, dashboard, API, command-line tool or notebook) **and** the code that created your submission files, not just one of them. At the top of your project, keep only `README.md`, `submission/`, `FULL_CODE/` and your `.gitignore` (plus a `LICENSE` file if you have one). The *Participant Coding Guidelines* (rule 12) list what your README must cover. If you built inside a pack folder, replace the pack's README with yours (rename the pack's one to `PACK_README.md` if you want to keep it).
- **Retail and Healthcare only:** a folder named exactly `submission` at the top of your project, next to your `README.md`, with the files below.
- **Do not include** the pack's data files (`data/`, and for B1 also `sql/`), downloaded models, `.env`, `.venv` or `node_modules`. Your README should say where to put the data and how to get any models.

ID	Files you must put in `submission/`
R1	`R1_predictions.csv`
R2	`R2_predictions.csv` and `R2_size_chart_top5.csv`
R3	`R3_predictions.jsonl`
R4	`R4_predictions.csv` (optional extra: `R4_cluster_allocation.csv`)
R5	`R5_predictions.csv`, `R5_emerging_flags.csv` and `detector_cmd.txt`
H1	`H1_predictions.csv`
H2	`H2_predictions.jsonl`
H3	`H3_predictions.csv`
H4	`H4_predictions.csv` and `H4_resolution.csv`
H5	`H5_answers.jsonl`, `H5_visit_list.csv` and `H5_footprint.json`
B1, B2	No `submission/` folder. There are no prediction files.

See the picture. Page 2 of the *Participant Coding Guidelines* shows this layout as an annotated picture, with examples for other stacks (web and mobile apps, Java, .NET, notebooks), a BFSI version, a list of what must never be in your zip, and a README skeleton.

3. Add anything else the "What to submit" section of your problem's README asks for. For example, **R1** needs a one-line command in your own README that reproduces your submission.

4. Make the zip from a fresh git clone. You need git; if you do not have it, install it from https://git-scm.com before the event. A fresh clone contains only what you committed. It leaves out everything listed in your `.gitignore`.

1. **Before you commit,** make sure your `.gitignore` contains these lines: `.env`, `data/`, `models/`, `.venv/`, `node_modules/` (B1 teams: also `sql/` and `*.db`). A file that was already committed stays in the clone. Also make sure `.gitignore` does not exclude your `submission/` files (for example with `*.csv` or `*.jsonl`).
2. Commit everything: run `git add -A`, then `git commit -m "final submission"`.
3. Open a terminal in the folder **above** your project folder (Step 5 shows how to open a terminal in a folder).
4. Make a clean copy named after your team and problem: run `git clone my-project T042_H3`. Replace `my-project` with your project folder's name (or your repository's web address), and `T042_H3` with your

own team and problem ID.

5. **Open the new `T042_H3` folder and check it.** You should see your `README.md`, `FULL_CODE/` with all your code and (Retail and Healthcare) `submission/` with your files. If something is missing, it was not committed or `.gitignore` excludes it. Fix that, commit again, delete the `T042_H3` folder and make the clean copy again.
6. Zip the `T042_H3` folder. You get `T042_H3.zip`.
   - **Windows:** right-click the folder → **Send to** → **Compressed (zipped) folder**. (On Windows 11, click **Show more options** first.)
   - **Mac:** right-click the folder → **Compress**.

5. Check before you submit:

☐ The zip is named `teamID_psID.zip` and contains `FULL_CODE/` with **all** your code (your demo app and the code that created your submission files) and **your own** `README.md`.

☐ You opened the cloned folder and saw everything you expected (step 4 above).

☐ No data files from the pack, no downloaded models, no `.venv`, no `node_modules`, no `.env`.

☐ Retail and Healthcare: `submission/` is at the top of your project, and file names are **exactly** as shown, with the same capital and small letters.

☐ Retail and Healthcare: your predictions are for the **test** data, not the training or dev data. They follow every rule in the "What to submit" section of your problem's README (for most problems, that means one row for every test ID).

☐ Retail and Healthcare: you ran `evaluate.py` on a training/dev prediction file made by **the same code**, and it worked without a format error.

If `evaluate.py` stops with an error message, **read the message**. It tells you exactly what is wrong with your file and how to fix it.

6. Upload your zip where and when the organisers tell you.

BFSI teams

For B1 and B2, build and show what the **published problem statement** asks for. Submit **all** your code in the same way (steps 1–6, without a `submission/` folder).

How you are judged

Retail and Healthcare: the score from `evaluate.py` on the hidden test data is **only 20%** of your marks.

What judges look at	Share of marks
Problem framing	20%
Technical soundness	25%
Score on hidden split (from `evaluate.py`)	20%
Real-world usability	15%
Responsibility	15%
Demo	5%

BFSI: there is no `evaluate.py` and no score file. The organisers will tell you how BFSI projects are judged.

Important notes

- **Retail and Healthcare:** do not try to get the hidden test answers by any trick. Your predictions must come from your own solution.
- **Healthcare teams:** All healthcare data is synthetic. No team may use real patient data. Every healthcare solution must be positioned as clinician-assisting, not clinician-replacing, and must include a safety/escalation path. Teams that present an autonomous diagnostic system without a human-in-the-loop will lose points on the Responsibility criterion. Medical information in these packs is simplified for this hackathon and is **not for clinical use**.
- **Which names are real?** The retailer ("Saffron Retail Group"), the hospital ("Chandrapuri District Hospital"), the bank ("Saffron Bank"), the insurer ("Saffron Life Insurance"), their brands, products and people are fictional. City and state names are real. **H2 and H4** use real medicine names (generic names and common Indian brand names) in a simplified form. Other healthcare packs use real generic medicine names or real ICD-10 diagnosis codes.
- **B1 teams:** India has no phone number range kept aside for fake numbers, so a mobile number in the B1 data could belong to a real person. **Never call or message any phone number in the data.**
- **B2 teams:** the insurance products, rates and numbers are invented. Nothing in the B2 data is financial advice.

Common problems and how to fix them

What you see	What to do
`'python' is not recognized…` (Windows)	Python is not installed, or "Add python.exe to PATH" was not ticked. Run the Python installer again and tick that box. You can also try `py` in place of `python`.
Typing `python` opens the Microsoft Store (Windows)	Install Python from https://www.python.org/downloads/ and tick "Add python.exe to PATH".
`command not found: python` (Mac or Linux)	Type `python3` in place of `python`.
`'pip' is not recognized` or `command not found: pip`	Use `python -m pip install …` (Mac or Linux: `python3 -m pip install …`) as shown in Step 4.
`error: externally-managed-environment` (Mac or Linux)	Make a private Python environment inside your problem folder: run `python3 -m venv .venv` then `source .venv/bin/activate`, and install the libraries again. Run `source .venv/bin/activate` again each time you open a new terminal. (Ubuntu: if `venv` fails, run `sudo apt install python3-venv` first.)
`ModuleNotFoundError: No module named 'pandas'` (or `numpy`, `pyarrow`)	Install that library as shown in Step 4.

What you see	What to do
`No such file or directory`, or `can't open file 'baseline/baseline.py'`	You are not inside your problem folder. Go back to Step 5 and check with `dir` or `ls`. On Windows, check that you are in the inner folder (see Step 2).
`evaluate.py` stops with an error message	Your file's format is wrong. The message says exactly what to fix (a column name, a missing ID, a wrong value). Fix it and run again.
My score is very low	That is normal for the baseline. Read **How you're scored** in your problem's README, then improve your solution.
I cannot find the test answers	That is on purpose. They are hidden, and the organisers use them to score your submission.
The zip will not open	The download may be broken. Download it again and check it with `SHA256SUMS.txt` (see Step 1).
I think there is a mistake in the data	Tell the organisers. If a file is corrected, a new version (for example `v1.1`) will be announced. Always use the latest version.

Words used in this guide

Word	Meaning
Problem statement (PS)	The challenge your team chose, such as R2 or H3.
Synthetic data	Data created by a computer program. It looks realistic but is not about real people.
Zip file	One compressed file that holds many files and folders. You "unzip" or "extract" it to get them out.
Terminal	A window where you type commands: Command Prompt on Windows, Terminal on Mac.
CSV / JSON / JSONL / Parquet	Data file formats. CSV is a table (opens in Excel). JSON and JSONL hold structured text (JSONL has one record per line). Parquet is a compressed table that pandas reads.
Training / dev data	Data **with** answers. Use it to build and check your solution.
Test data	Data **without** answers. You make predictions for it, and those predictions are scored.
Label	The correct answer for a row of data.
Baseline	A simple starting solution that already works. Try to beat it.
`evaluate.py`	The scoring program. It checks your file's format and gives a score.
Primary score	The main score from `evaluate.py` (`primary_score` in its `.json` file). Higher is better.

Licence

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